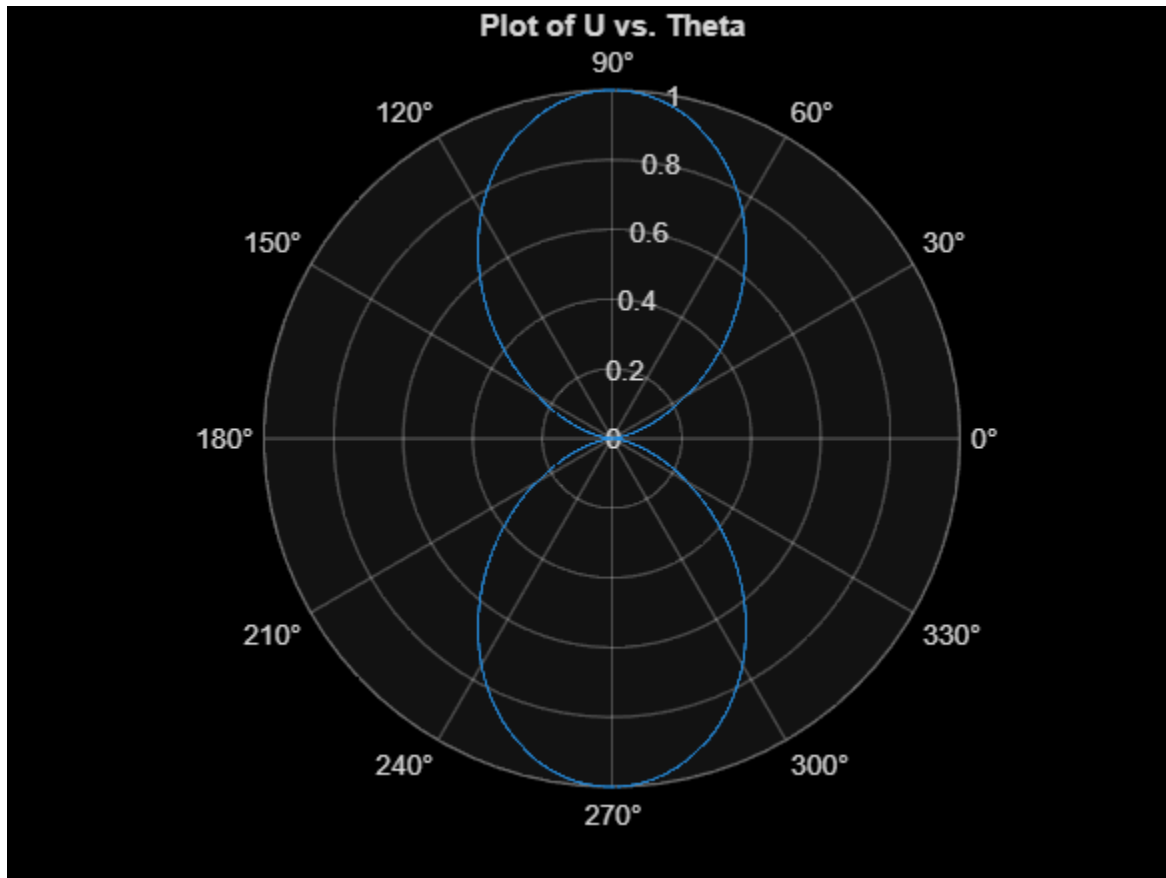

part a)

```
theta = linspace(0,2*pi, 10000);  
U = (sin(theta)).^2;  
polarplot(theta, U);  
title('Plot of U vs. Theta');  
grid on;
```



part b

```
theta = linspace(0, pi, 10000);  
integrand = (sin(theta)).^3;  
Prad = 2*pi * trapz(theta, integrand);  
Dmax = 4*pi*max(U)/Prad;  
disp(Dmax);  
  
%Identifying the direction of maximum radiation  
theta = linspace(0,2*pi, 10000);  
theta_max = theta(U == max(U));  
disp(rad2deg(theta_max(1)));  
  
1.5000
```

90.0090

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